Reconstructed rules in a synthetic Swift payment book

Ten million fictitious payments, 30 April to 31 July 2023. This deck states the business rules the file actually follows, one feature group per slide, with the chart that shows the rule and the live-market behaviour that pattern would represent.

ONE SLIDE EACH

1 Data quality and scale

2 Time patterns

3 Currency

4 Payment value

5 Routing and corridors

6 Path structure

7 Duration

8 Party types

9 Accounts and tokens

10 Anomaly labels

Full population, n = 10,000,000. 30 Apr 2023 – 31 Jul 2023, UTC.

How a pattern became a rule

1 Ingest out of core

DuckDB scans the 4.6 GB CSV. The raw file is never loaded into pandas. Columns arrive as text, then amounts, counts, and the timestamp are cast.

2 Keep a real payment row

A row stays when EPOCH_SECONDS parses as a number. Hour, weekday, and day are extracted in UTC. Domestic means the ordering and beneficiary BIC share a country. Path length counts BIC steps. A path gap is the token ‘…’.

3 Aggregate the full population

Each feature group is a SQL aggregation. Rates and medians use all 10 million rows. Histogram shapes use a sample drawn after the filter that defines the group. Those samples are not used for the sizes stated here.

4 Accept only a measured effect

A rule states a condition, an effect, a size, and a scope. It is kept when the chart shows that size on the full file. Where a starting idea disagreed with the chart, the slide states the measured pattern.

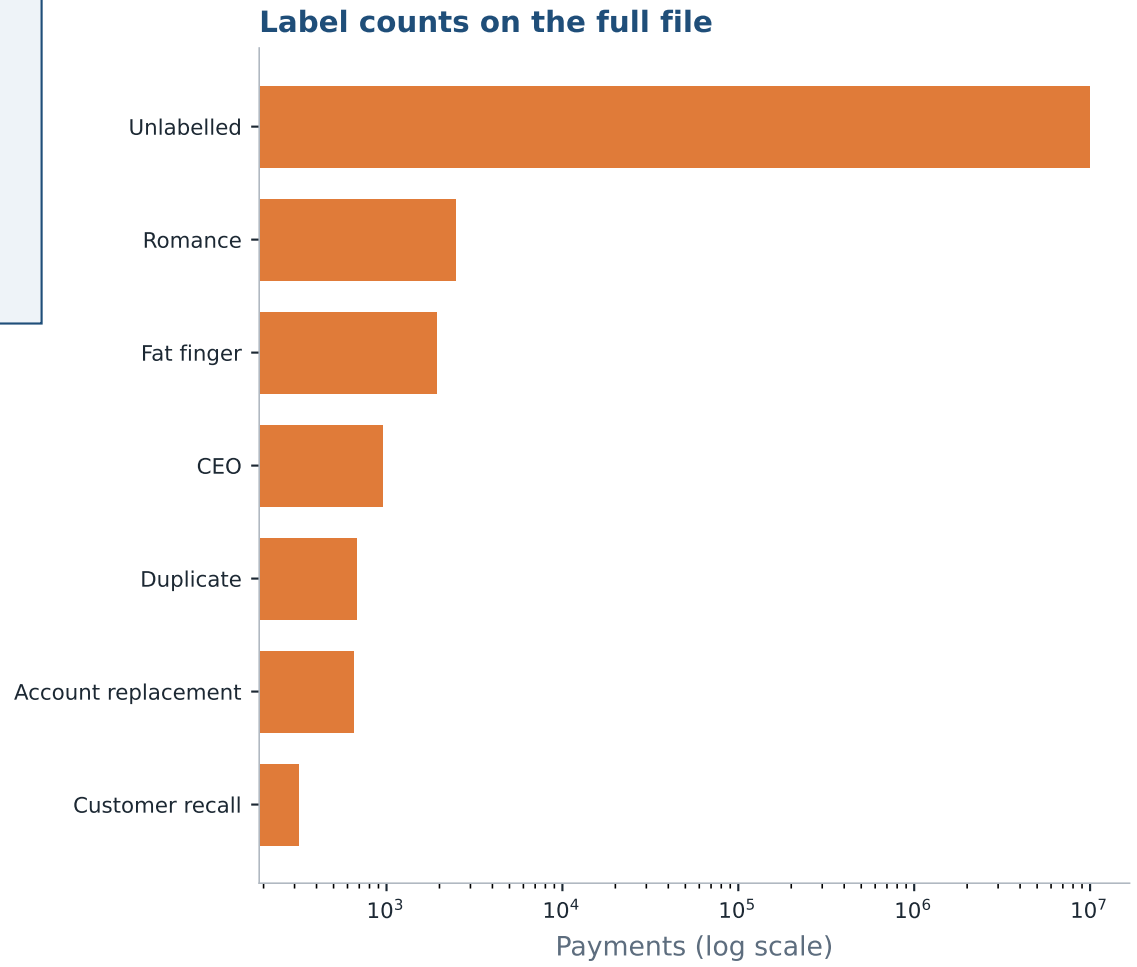
The book is complete, and labels are rare

RECONSTRUCTED RULE

On every retained row, timestamp, USD amount, BIC path, and both detail fields are populated, and the USD amount is positive. Exact-duplicate rows are 1 in 10 million. An anomaly label is present on 7,006 payments, which is 0.070 percent. Scope: the full retained file, 30 April 2023 11:22 UTC through 31 July 2023 09:56 UTC.

IF THIS WERE A LIVE PAYMENT SYSTEM

A monitoring team can treat the file as the population, not a sample with a junk tail. Any screen has to work at about one labelled payment in 1,400. The warning text lives in a column of the first payment. It is not a second pile of broken rows.



Same counts: dashboard page 1, upper right.

Full population, n = 10,000,000. 30 Apr 2023 – 31 Jul 2023, UTC.

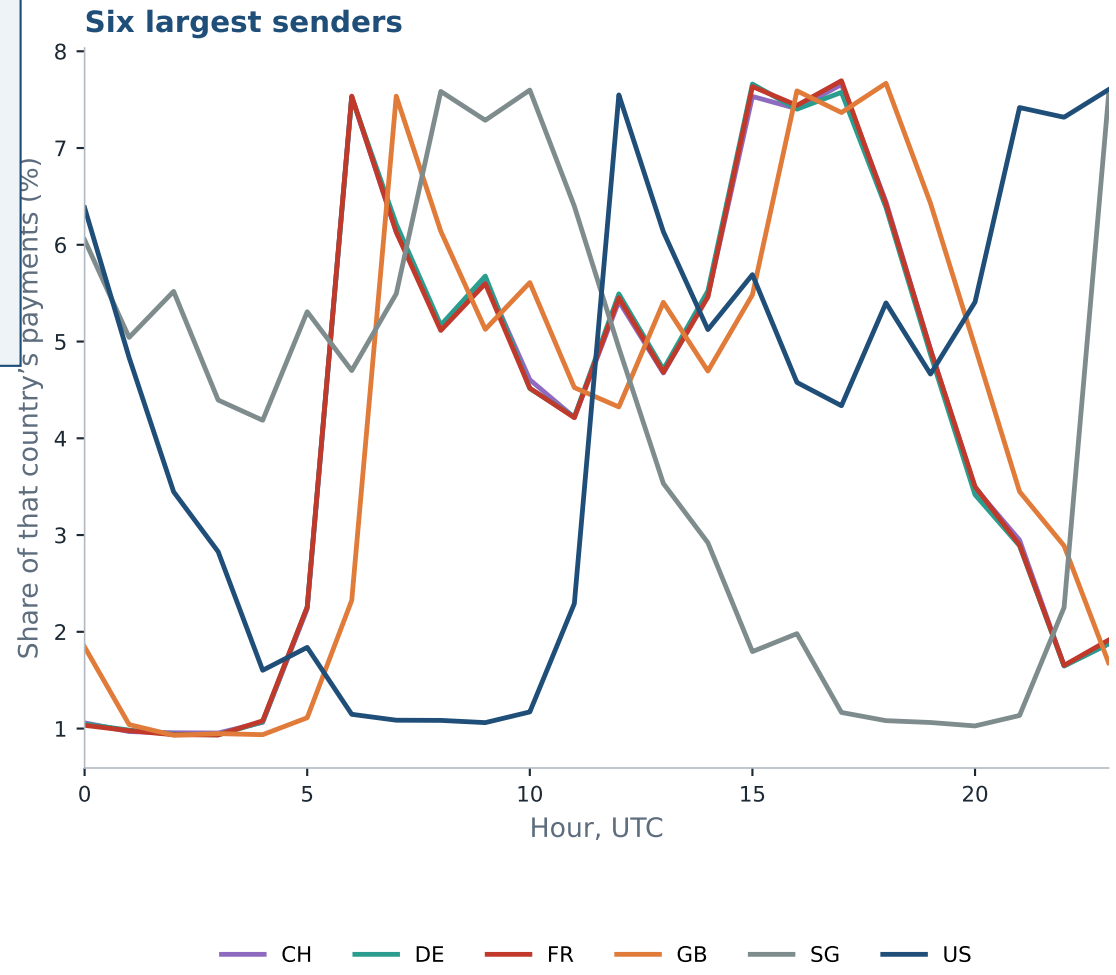
The book moves west through the UTC day

RECONSTRUCTED RULE

The UTC hour with the most payments shifts with the sender’s country: China 08:00, Japan 09:00, Singapore 10:00, Germany 15:00, France and Switzerland 17:00, Britain 18:00, the United States 23:00. A weekday carries 1.70–1.83 million payments. Saturday carries 0.43 million and Sunday 0.68 million. The label rate stays between 0.056 percent and 0.089 percent in every UTC hour. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

People pay during their own working day, so Asian banks fill the UTC morning, European banks the UTC afternoon, and US banks the late UTC evening. The weekend book is about one third of a weekday. An after-hours screen sees the same label rate as midday.



Weekday bars and the hourly label rate: dashboard page 2.

Full population, n = 10,000,000. 30 Apr 2023 – 31 Jul 2023, UTC.

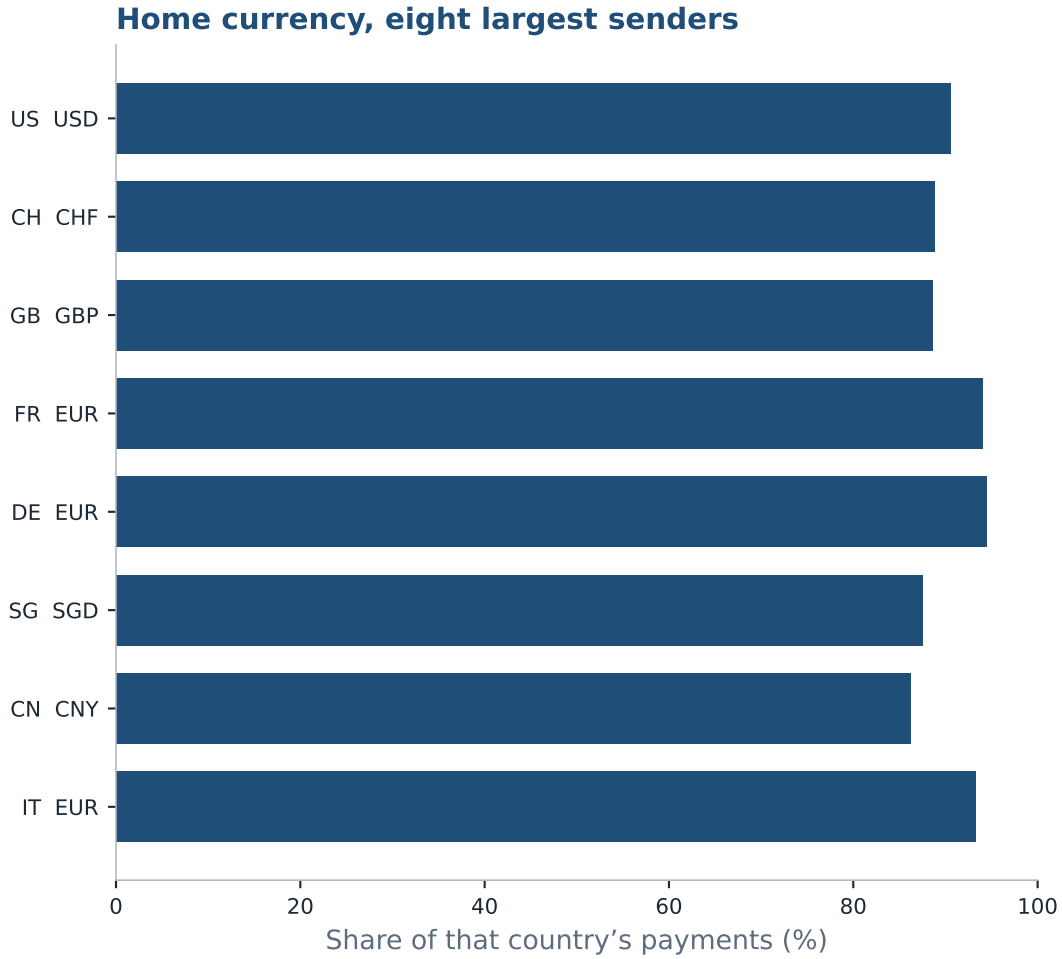
The sender’s country picks the currency

RECONSTRUCTED RULE

For each of the 12 largest sending countries, 86–95 percent of payments use that country’s home currency: the euro in France, Germany, Italy, Belgium, Luxembourg, and the Netherlands; the dollar for 90.5 percent of US senders; the franc 88.8 percent; sterling 88.6 percent; the Singapore dollar 87.5 percent; the yuan 86.2 percent; the Hong Kong dollar 88.3 percent. A cross-border payment is converted 93 percent of the time. A domestic payment is converted 37 percent of the time. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

Firms and people pay in the currency of the bank that sends, which is how a local clearing currency becomes the default. Crossing a border usually means an FX ticket. Staying inside one country still converts more than a third of the time, so a domestic flag is a weak substitute for an FX flag.



Domestic versus cross-border FX: dashboard page 3, lower right.

Full population, n = 10,000,000. 30 Apr 2023 – 31 Jul 2023, UTC.

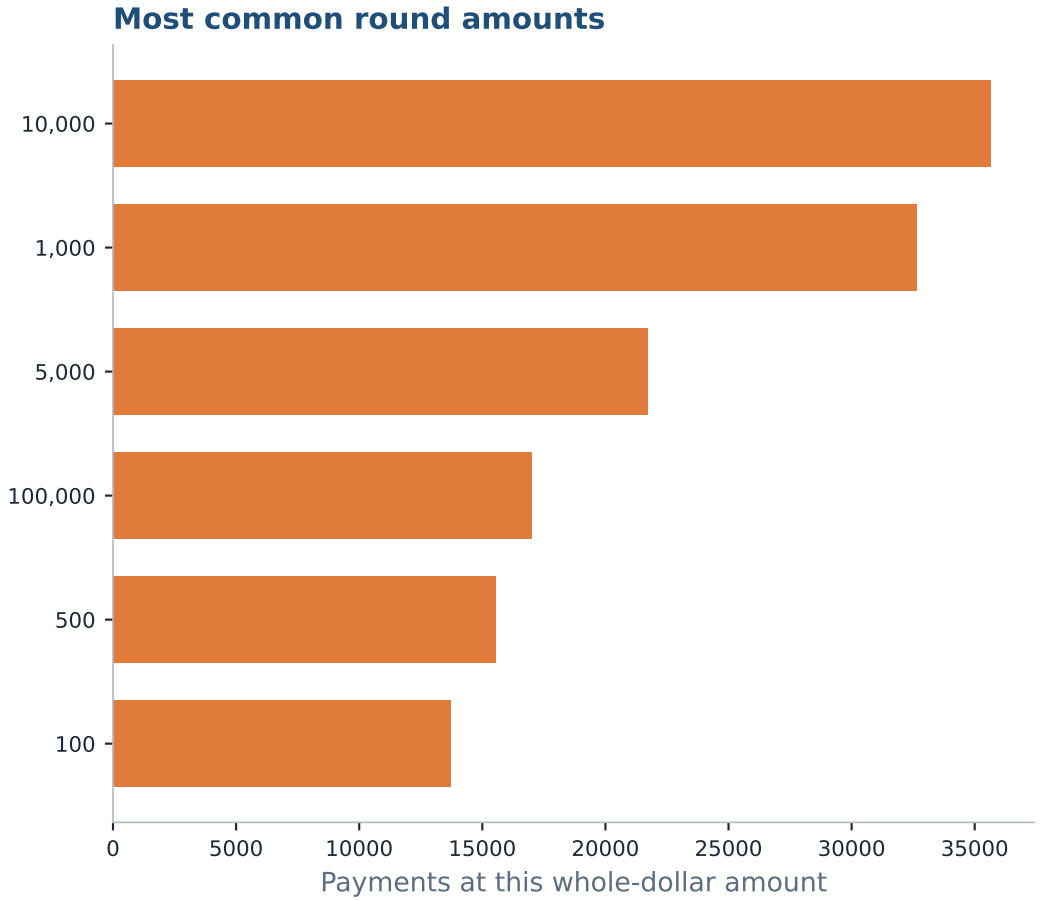
Amounts sit on a log scale, and round numbers stick

RECONSTRUCTED RULE

log10 of the USD amount is a single hump, the shape of a log-normal book. Among whole-dollar amounts from 100 to 100,000, the most common values are 10,000 (35,650 payments), 1,000 (32,630), 5,000 (21,709), and 100,000 (16,995). The level of the amount follows who is paying: both parties business, median 22,888 dollars; either side a consumer, about 3,200 dollars; both consumers, 463 dollars. Scope: all payments; the shape is the full-file histogram on dashboard page 4.

IF THIS WERE A LIVE PAYMENT SYSTEM

Payment size multiplies rather than adds: a typical firm payment and a typical household payment live an order of magnitude apart. Round amounts are price points and invoice conventions, the same way a cash economy piles up on notes of 100 and 1,000.



Log-amount shape and party-type histograms: dashboard page 4.

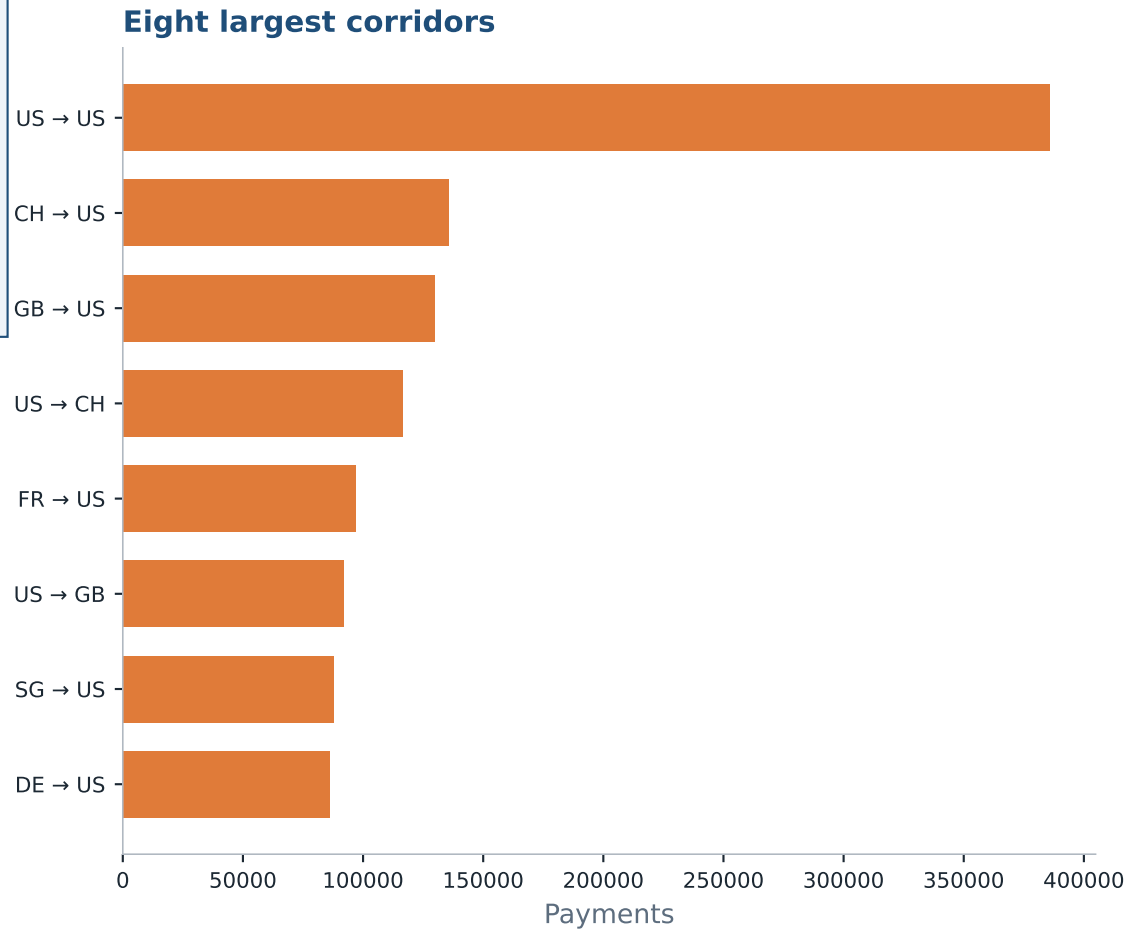
Traffic hubs through the United States

RECONSTRUCTED RULE

The file contains 26,764 corridors. The largest, United States to United States, is 385,598 payments, 3.9 percent of the book. The top 15 corridors are 16.1 percent. The most repeated bank pair, 0001MQM0 to 0009TNT0, occurs 20,005 times. Sender and beneficiary banks share a country on 24 percent of US payments and on under 10 percent for every other large sender. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

This is a correspondent network with a dollar hub, plus a large domestic American book. Bank pairs do repeat, which is what a sticky relationship looks like, and the long tail of 26,764 corridors is what an open network looks like. No single pair is the market.



Heatmap, BIC pairs, and domestic share: dashboard page 5.

The product sets the path, and the path sets the FX

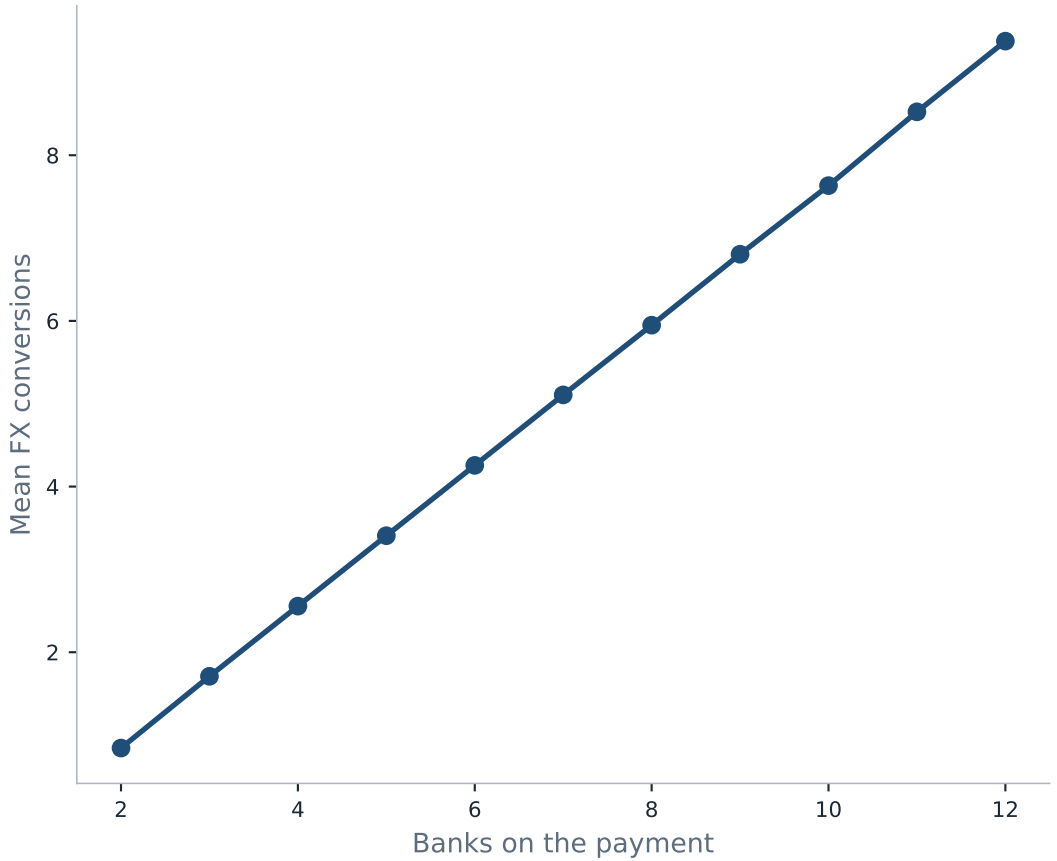
RECONSTRUCTED RULE

Direct is exactly two participants: 6,323,280 payments, every Direct payment, and no Covered or Indirect payment has two. Path length equals the participant count on 92.3 percent of payments, and equals that count plus the gap token on 99.4 percent. CONTINUITY_QFLAG is N exactly when the path contains '…': 767,204 payments, and zero rows disagree. Mean FX conversions rise by about 0.85 for each bank added, from 0.84 at two banks. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

Direct is a two-bank product. Covered and Indirect are the same payment handed along a chain. Each extra bank is another place a currency can change, so the FX count walks up almost one for one with the chain. A '…' in the path is a missing link in that chain, and the continuity flag is that fact written twice.

One more bank, almost one more conversion



Participant mix, path heatmap, continuity table: dashboard page 6.

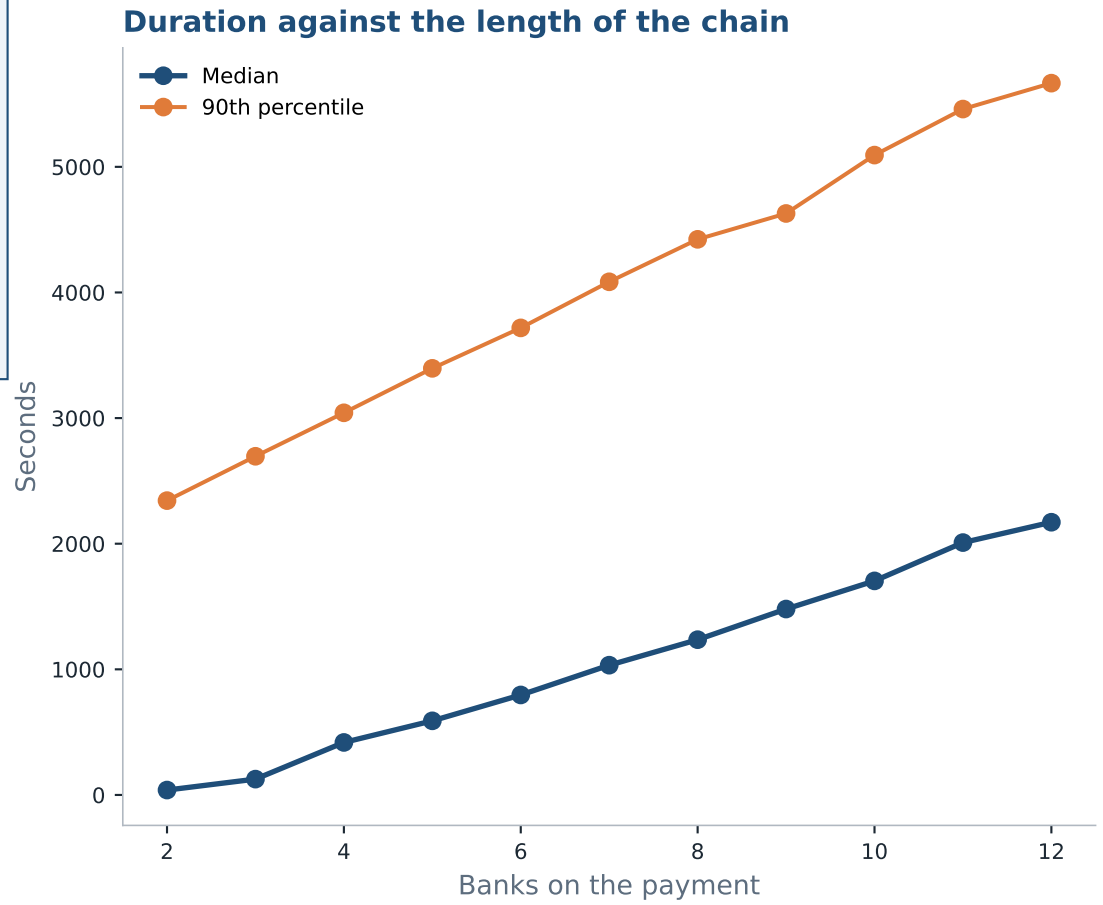
Each extra bank adds waiting time

RECONSTRUCTED RULE

Median duration is 39 seconds for Direct, 281 seconds for Indirect, and 383 seconds for Covered. It rises with the chain: 39 seconds at two banks, 126 at three, 418 at four, 1,704 at ten, and 2,171 at twelve. A path that contains '…' has median 155 seconds; a continuous path has median 54 seconds. The cross-border median is 55 seconds and the domestic median is 167 seconds. Scope: all payments. The far tail, past twelve banks, is a few dozen payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

Duration here is processing and hand-off time, not the length of a flight. A direct payment clears in under a minute at the median. A covered or indirect payment waits through extra banks. A gap in the path is a repair. Crossing a border, on its own, is not the slow case in this file.



Duration by product and by path gap: dashboard page 7.

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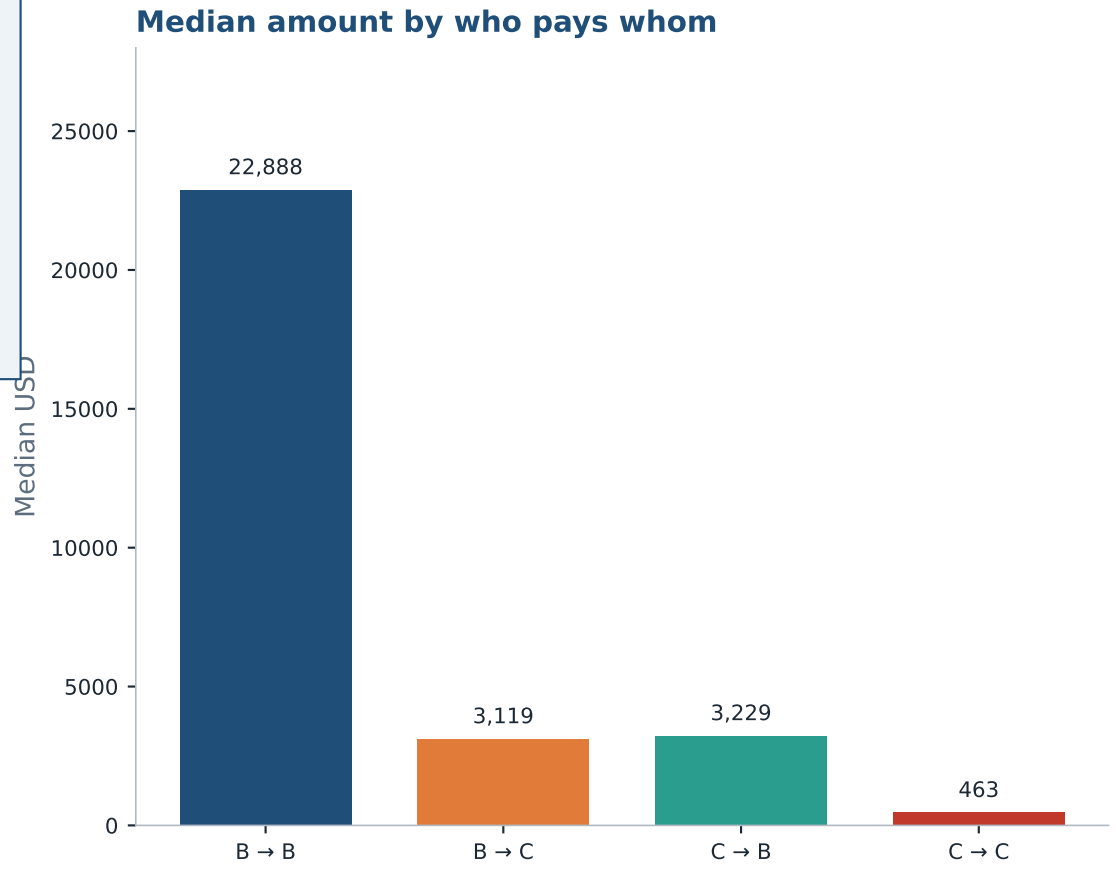
Households pay hundreds. Firms pay tens of thousands.

RECONSTRUCTED RULE

Business-to-consumer is the modal flow, 5.60 million payments, median 3,119 dollars. Business-to-business is 2.27 million payments, median 22,888 dollars. Consumer-to-business is 0.61 million, median 3,229 dollars. Consumer-to-consumer is 1.51 million, median 463 dollars. The sender is a business on 75–86 percent of payments in every large country, and that share is about four fifths for Direct, Covered, and Indirect alike. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

Consumers move household money. Firms move invoices and treasury. The routing product does not pick the customer, and the sender’s country does not either: every large market in this file is a business-heavy book inside a narrow band.



Counts, country mix, and routing mix: dashboard page 8.

Europe carries an IBAN. The active accounts are few.

RECONSTRUCTED RULE

There are 391,132 ordering accounts. A third are used once, the median account sends 3 payments, and the busiest sends 75,803. The top 1 percent of ordering accounts send 46 percent of payments. Of 2.59 million beneficiary accounts, 78 percent receive a single payment. An IBAN token is on 91–96 percent of payments from Switzerland, Britain, and the euro area, and on 0 percent from the US, Singapore, China, Hong Kong, Japan, Australia, and the Isle of Man. Ordering details contain the BIC country code on 97–100 percent of European payments and on 67–78 percent from the US, Singapore, China, and Hong Kong. Scope: all payments.

IF THIS WERE A LIVE PAYMENT SYSTEM

European rails in this file identify the account with an IBAN. US and Asian rails do not. Origination is a wholesale pattern: many quiet accounts, and a small set that sends nearly half the book. Beneficiary accounts look like invoices, opened for one payment.

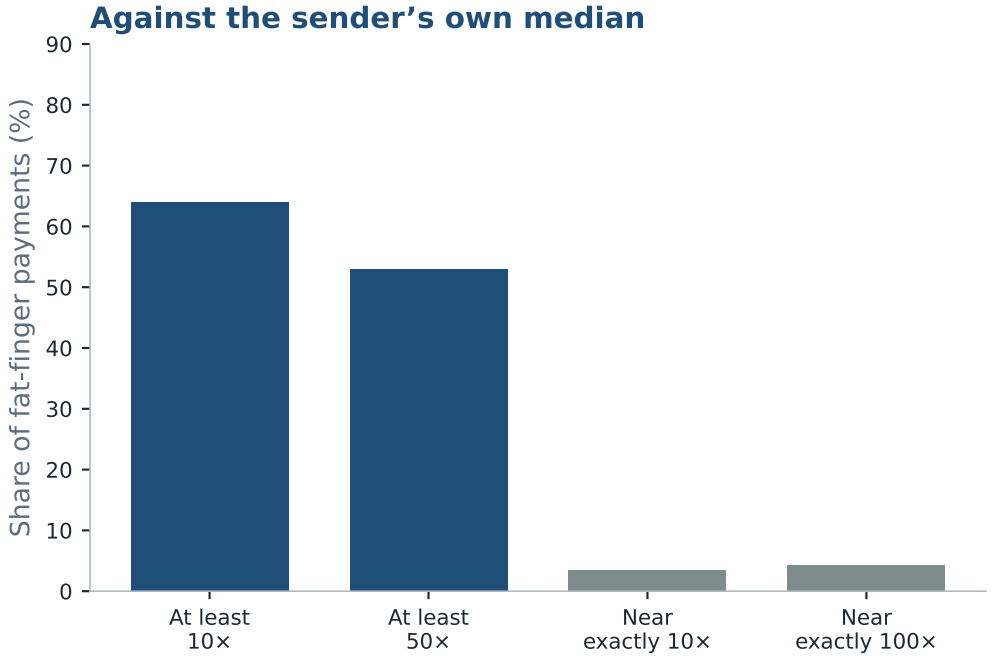
What the sender’s text reveals



Payments-per-account histograms: dashboard page 9.

Four labels have a signature. Two do not.

Romance	**2,476**	Always a consumer sender. 73% pay a consumer. Median $109. Median of 6 payments to the same beneficiary. 97% cross-border.
Fat finger	**1,940**	Median is 71× what that ordering account usually sends. 64% are at least 10×. 53% are at least 50×. A narrow band around exactly 10× holds 3.5%, and around 100× holds 4.3%.
CEO	**945**	Always a business sender. 99% cross-border. Median $20,602, beside the ordinary business-to-business median of $22,883.
Duplicate	**679**	The same two accounts, amount, and currency return after a median of 18 minutes. Other repeats of that key return after 15 days.
Account replacement	**649**	99.4% cross-border. A first-seen beneficiary account is 18% of these payments and 21% of ordinary payments.
Customer recall	**317**	Median $3,164 and 78% business senders, in line with the ordinary book. No separate signature in amount, party, or corridor.



In a live system this is an extra keystroke on a familiar account, a double submit in one sitting, and a repeated small personal transfer abroad.

Duplicate-gap histogram: dashboard page 10.

What an investigator would actually screen on

Amount versus the sender’s own history
Flag a payment near 50 times that ordering account’s median, and wider. A fixed dollar cutoff misses a 71× fat finger on a small account and catches ordinary business payments.

A short duplicate window
Same ordering account, beneficiary account, amount, and currency inside about 20 minutes. A window of days mixes the labelled duplicates with ordinary repeats.

Romance as a repeated personal flow
A consumer sender, a small amount, a cross-border beneficiary who already received several of these payments.

Leave the clock out of the fraud score
The label rate is flat across the UTC day. Hour of day describes where the sender sits, which is useful for operations and useless as a suspicion score in this file.

Read FX off the conversion count
Domestic payments still convert 37 percent of the time. The conversion count, and the length of the bank chain, are the FX rules.

Treat path and duration as operations
Participant count, the ‘…’ token, and duration move together. Labelled payments sit near 2.5 banks, the same neighbourhood as ordinary Direct traffic, so the chain is a service clock rather than a fraud feature.

What these rules are, and what they are not

The file is synthetic.
Swift states that the payments were generated from hard-coded rules, NumPy draws, and the Faker library. The rules above describe this file. The economic readings are what the same pattern would mean in a live system.

Small labels stay thin.
Customer recall has 317 payments and account replacement has 649. A signature that is absent at that size may be absent because the generator did not write one, or because these fields cannot see it.

Two familiar stories are not in the columns we tested.
A CEO payment is not larger than an ordinary business-to-business payment. A first-seen beneficiary account is about as common on account-replacement payments as on ordinary ones. A ‘new beneficiary’ test beyond that pair was not run.

Samples never carry a size.
Every percentage, median, and count in this deck is a full-population aggregate. Samples were used only to draw distribution shapes on the dashboard.